

Screening of Coulombic Interactions To Achieve a Higher Power Factor in Conjugated Polymers

Pawan Kumar,^{*,†} Anas Abutaha,[†] Gang Wu,[†] Madeleine P. Gordon, Jose Recatala-Gomez, Jeffrey J. Urban, and Kedar Hippalgaonkar^{*}



Cite This: *ACS Appl. Mater. Interfaces* 2025, 17, 9431–9439



Read Online

ACCESS |



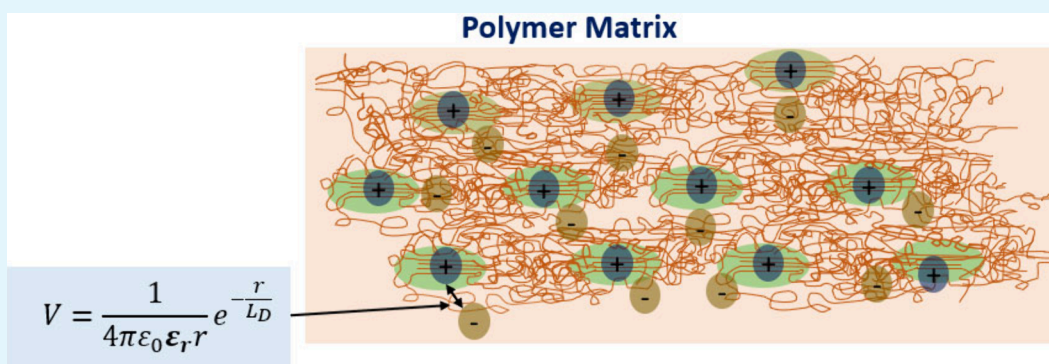
Metrics & More



Article Recommendations



Supporting Information



ABSTRACT: Thermoelectric properties of conducting polymers typically suffer from molecular chain disordering, as charge transport is predominantly controlled by morphology. This is especially more problematic when counterions are introduced to tune the carrier concentration for optimal thermoelectric performance, which disturbs the morphology further. In this work, we introduce a new avenue for enhancing thermoelectric properties without needing to regulate the morphology, namely, by controlling the coulombic interaction between polarons and counterions. We perform *in situ* de-doping thermoelectric experiments over 3 orders of magnitude change in electrical conductivity of three distinct thermoelectric polymers, namely, poly(3-hexylthiophene-2,5-diyl) (P3HT), poly[2,5-bis(3-dodecylthiophen-2-yl)thieno[3,2-*b*]thiophene] (PBTTT-*C*₁₂), and poly[2,5-(2-octyldodecyl)-3,6-diketopyrrolopyrrole-*alt*-5,5-(2,5-di(thien-2-yl)thieno[3,2-*b*]thiophene)] (OD-PDPP2T-TT) conjugated polymers, followed by grazing-incidence wide-angle X-ray scattering (GIWAXS) to study their respective morphologies. We demonstrate a 9-fold enhancement in the thermoelectric power factor in OD-PDPP2T-TT compared to PBTTT-*C*₁₂ and link it to the coulombic screening of charge carriers, including in the optimally doped regime. We support this hypothesis using Boltzmann transport equations and show that, in both P3HT and PBTTT-*C*₁₂, as the polymer is doped, impurity scattering remains the dominant scattering mechanism, while in OD-PDPP2T-TT, the scattering mechanism changes from impurity to acoustic phonon limited, resulting in more effective screening of ionized counterions. Our results provide an additional knob to enhance the fundamental understanding of thermoelectric physics of conducting polymers and provide a pathway to achieve higher performance in the field of organic thermoelectrics.

KEYWORDS: organic thermoelectric, dielectric constant, doping, power factor, conducting polymer, thin film

INTRODUCTION

A large part (~66%) of waste heat due to industrial processes is released below 200 °C,¹ and thermoelectric devices possess considerable potential for converting it into useful electrical power. The efficiency of a thermoelectric material depends upon the figure of merit, $ZT = \frac{S^2\sigma T}{\kappa}$, where S is the Seebeck coefficient, σ is the electrical conductivity, κ is the total thermal conductivity, and T is the thermodynamic temperature. Although inorganic materials, the commercially available industry standard, have shown good thermoelectric properties over a wide range of operation temperatures, they are expensive and potentially toxic, which ultimately limits their widespread deployability.^{2,3} On the other hand, conducting

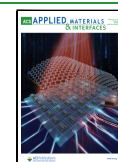
polymers are good contenders for near-room-temperature applications due to their solution processability and earth abundance. Recently, poly(ethylenedioxythiophene) (PEDOT)-based polymers have shown promising thermoelectric properties at room temperature by either controlling the morphology or dopant volume.^{4–6} Along with PEDOT,

Received: November 28, 2024

Revised: January 14, 2025

Accepted: January 21, 2025

Published: January 30, 2025



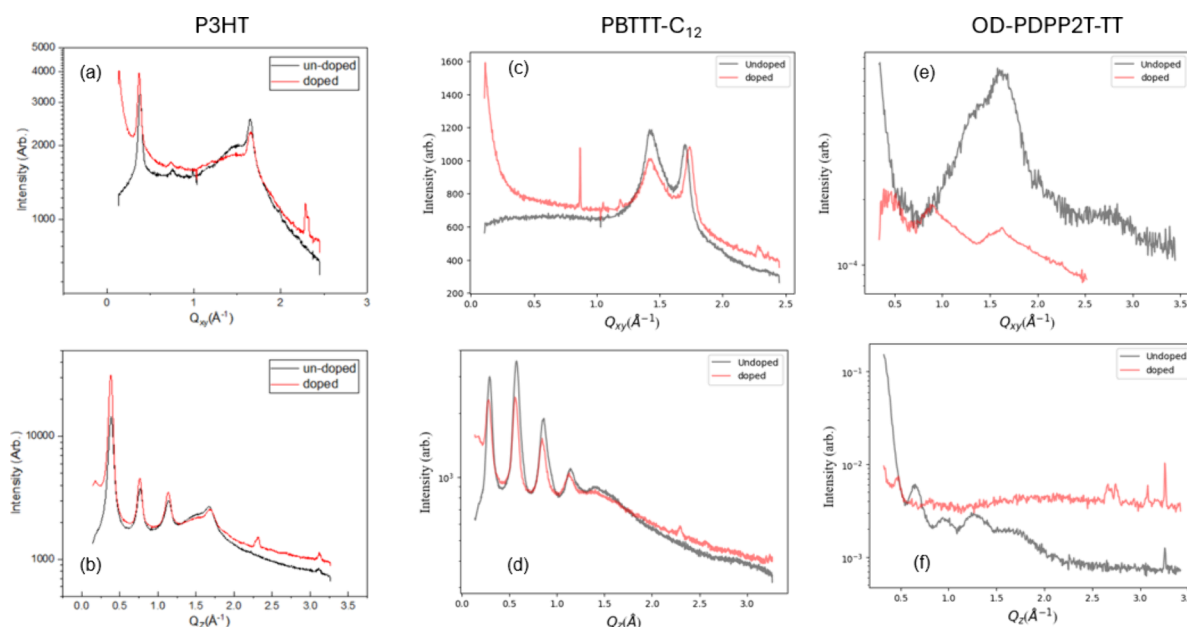


Figure 1. (a–f) GIWAXS data of undoped and doped P3HT and PBTTT- C_{12} and GIXRD OD-PDPP2T-TT polymer films on the SiO_2 substrate. Two-dimensional (2D) diffractograms of the P3HT, PBTTT, and OD-PDPP2T-TT thin films on the SiO_2 substrate before (black) and after (red) doping of P3HT, PBTTT, and OD-PDPP2T-TT thin films.

poly(3-hexylthiophene) (P3HT) and poly(2,5-bis(3-alkylthiophen-2-yl)thieno[3,2-*b*]thiophene) (PBTTT) have also shown promising thermoelectric (TE) properties at room temperature by modifying their structural morphology^{7,8} and controlling the doping process to avoid further structural disorder due to fluctuation in the interchain spacing known as paracrystallinity.^{9–13} The dopants introduce polarons within the polymer backbone, which provide itinerant charge carriers, resulting in electrical transport. However, with the introduction of polarons via such externally added dopant molecules, morphological changes and increased disorder can result depending upon their size and distribution. These disruptions can be detrimental to conductivity^{7,13–16} and arise from the distortion due to ionized counterions in the polymer matrix. In addition, a strong coulombic interaction between ionized counterions and polarons effectively hinders the charge mobility of the polarons further. Aubry et al.¹⁷ have controlled the separation between counterions and polarons by choosing dodecaborane (DDB)-based dopants that act to reduce the coulombic interaction between them and, hence, improve the conductivity of the P3HT polymer film. However, due to the large size of the DDB- F_{72} dopant, morphological disorder is increased, which adversely impacts the charge mobility. Therefore, we hypothesize that reducing the coulombic interaction between counterions and delocalized polarons without disturbing the morphology will be the key to further enhance the TE properties in conducting polymers. This can effectively be done by either controlling the dielectric environment or distance between the polarons and counterions.

The effect of the dielectric constant on photovoltaic properties of organic polymers has been well-studied.¹⁸ Cho et al.¹⁹ have shown that, by incorporation of polar nitrile side chains in a diketopyrrolopyrrole indacenodithiophene polymer (DPP-IDT-CN) compared to an analogous polymer with alkyl side chains (PIDT-DPP-alkyl), the dielectric constant of the donor material increases from 3.9 to 5. This increase results in

longer carrier lifetimes, improving both the open-circuit voltage and short-circuit current in DPP-IDT-CN/ C_{60} bulk heterojunction (BHJ) photovoltaic devices. Brebels et al.²⁰ have shown that, by replacement of alkyl side chains with oligo(ethylene glycol) side chains in a poly[2,6-(4,4'-dihexyl-4*H*-cyclopenta[2,1-*b*;3,4-*b'*]dithiophene-*alt*-*N*-alkylthieno[3,4-*c*]pyrrole-4,6-dione (PCPDT-TPD) copolymer, the dielectric constant changes from 3.1 to 6.3. Wang et al. have shown a remarkable enhancement in the dielectric constant in P3HT by adding sulfinyl and sulfonyl groups within the alkyl side chains.²¹ The dielectric constant at room temperature was shown to increase from 3.75 for P3HT to 7.4 for the sulfinylated P3HT polymers and 8.1–9.3 for the sulfonylated P3HT polymers. However, while side-chain modification is a powerful approach for improving the dielectric constant in P3HT, it often disrupts packing morphology and increases the distance between the π - π stacks, both of which are not beneficial for charge transport.^{22,23} Alternatively, a diketopyrrolopyrrole (DPP)-based low-bandgap poly[2,5-(2-decyltetradecyl)-3,6-diketopyrrolopyrrole-*alt*-5,5'-(2,5-di(thien-2-yl)thieno[3,2-*b*]thiophene)] (DT-PDPP2T-TT) copolymer has also shown a higher dielectric constant without side-chain modification.²⁴

In this work, we therefore show that the dielectric constant of the polymer matrix itself can be used to boost TE properties without causing major morphological disruptions at various levels of doping. Here, we reveal the connection between the dielectric constant and thermoelectric transport properties of p-doped P3HT, PBTTT, and poly[2,5-(2-octyldodecyl)-3,6-diketopyrrolopyrrole-*alt*-5,5'-(2,5-di(thien-2-yl)thieno[3,2-*b*]thiophene)] (OD-PDPP2T-TT) copolymers. P3HT and PBTTT- C_{12} are well-known p-type semi-crystalline polymers.^{25,26} OD-PDPP2T-TT, which is a donor–acceptor (D–A) copolymer has also shown p-type nature with similar degrees of semi-crystallinity.²⁷ We measure the thermoelectric power factor ($PF = S^2\sigma$) and compare the effects of the degree of disorder in these conducting polymers. We have measured

an optimum PF of $66 \mu\text{W m}^{-1} \text{K}^{-2}$ for OD-PDPP2T-TT, which is >8 times larger than PBTTT, which has a PF = $7.8 \mu\text{W m}^{-1} \text{K}^{-2}$, and >40 times higher than P3HT, which shows PF = $1.6 \mu\text{W m}^{-1} \text{K}^{-2}$, with similar or higher structural disorder described by their paracrystallinity.¹² The Seebeck coefficient only depends upon the carrier concentration (n) (to fix the effective density of state mass), while the conductivity (σ) depends upon both the carrier concentration (n) and carrier mobility (μ). Hence, with the increase of the dielectric constant to aid screening of the polarons in OD-PDPP2T-TT, their coulombic interaction with the externally introduced counterions can be reduced, thus enhancing the charge mobility and resulting in an improved σ .

RESULTS AND DISCUSSION

First, we study the morphology in both pristine and doped films of P3HT, PBTTT- C_{12} , and OD-PDPP2T-TT polymers by performing grazing-incidence wide-angle X-ray scattering (GIWAXS) and grazing-incidence X-ray diffraction (GIXRD) for both pristine and doped films, as shown in panels a–f of Figure 1. All films were doped with a 0.03 M concentration of FeCl_3 . For P3HT, out-of-plane ($h00$) scattering peaks up to the third order attributed to the lamellar structure as well as an interchain (010) peak attributed to π – π stacking are observed, indicating a semi-crystalline nature with preferential edge-on orientation.¹⁰ For PBTTT (panels c and d of Figure 1), peaks indicative of the lamellar structure up to the fourth order in the out-of-plane direction as well as an interchain peak were observed, revealing an edge-on orientation with even better chain alignment than P3HT, as previously reported in the literature.²⁸ In the case of OD-PDPP2T-TT (panels e and f of Figure 1), while lamellar peaks are observed in the out-of-plane direction, the interchain peak is prominent in the in-plane direction, suggesting that PDPP2T-TT thin films are edge-on-oriented with respect to the SiO_2 substrate. For pristine P3HT, the (100) and (010) distances were found to be 15.85 and 3.80 Å, respectively, which are close to values reported in the literature.¹⁰ After doping, P3HT films retain their edge-on orientation, and no appreciable shift in (010) peak location is seen, as shown in panels a and b of Figure 1, suggesting that the counterions do not change the π – π stacking of the film further. In the case of pristine PBTTT, lamellar and π – π stacking distances were found to be 20.65 and 3.678 Å, respectively, which are consistent with reported values.²⁹ After doping, the peak positions shifted to 21.43 and 3.61 Å and edge-on orientation was maintained. The small shift in lamellar and π – π stacking distances shows a marginal variation in structural disorder of PBTTT with FeCl_3 doping. The shift in lamellar distance indicates that the counterions are intercalated between the alkyl side chains and concurrently reduce the π – π stacking distance, which, in turn, strengthens the π – π orbital overlap, hence improving the delocalization of charge carriers.^{10,30} For OD-PDPP2T-TT, the lamellar and π – π stacking distances were observed to be 19.3 and 3.87 Å, respectively, which are comparable to those published previously.²⁷ After doping, the lamellar distance increases and reaches 23.2 Å, while no change is observed in the π – π stacking distances, and it stays at 3.92 Å.

The structural disorder is generally described by paracrystallinity (g), which represents the fluctuation in the π – π stacking spacing distance. A general relationship between charge transport and paracrystallinity in conducting polymers is known,¹² where higher g is known to produce more localized

states in a material's electronic band gap, which limits charge transport.¹² Here, paracrystallinity [$g_{(010)}$] can be derived from $g_{(010)}^2 = \langle (\delta/d_0)^2 \rangle = \langle d^2 \rangle / (d_0^2) - 1$, where d is the π – π stacking distance observed from the radial intensity plots from Figure 1b and $d_0 = \langle d \rangle$ is the averaged π – π stacking distance along the (010) direction, while $\delta = d - d_0$ represents the change of the π – π stacking distance between the adjacent polymer chains in that specific crystal direction of [010]. In reciprocal space, g can be written as $g_{(010)} = \sqrt{\frac{\Delta q}{2\pi q_0}}$, where q_0 is

the peak position and Δq is the full width at half maximum (fwhm) of the scattering peak^{12,18} along the π – π stacking direction in the GIWAXS measurement. The π – π stacking distance was chosen because it is the predominant direction for charge transport. As shown in Table 1, the pristine PBTTT

Table 1. Paracrystallinity Change from Pristine to Doped P3HT, PBTTT, and OD-PDPP2T-TT Polymer Thin Films

material	q_0 (Å ^{−1})	Δq (Å ^{−1})	$g_{(010)} = \sqrt{\frac{\Delta q}{2\pi q_0}}$ (%)
P3HT (undoped)	1.6533	0.0737	8.4
P3HT (doped)	1.657	0.07908	8.7
PBTTT (undoped)	1.705	0.0471	6.6
PBTTT (doped)	1.7402	0.07018	8.0
OD-PDPP2T-TT (undoped)	1.62	0.2	13.8
OD-PDPP2T-TT (doped)	1.615	0.22	14.1

film possesses a lower g compared to P3HT, indicating that PBTTT chains tend to be more aligned under the same processing conditions. The OD-PDPP2T-TT film exhibits the highest g value compared to P3HT and PBTTT films. After doping, structural disorder was less affected in P3HT and OD-PDPP2T-TT films by FeCl_3 doping, but an appreciable effect was observed in PBTTT films as the g value increased from 6.6 to 8%.

While GIWAXS provides information about the structural morphology of the polymer chains, we next perform ultraviolet–visible (UV–vis) light absorption studies on the pristine and doped P3HT, PBTTT, and OD-PDPP2T-TT films to understand the extent of delocalization of the polarons. The spectra of P3HT thin films in Figure 2a show a dominant peak at $E = 2.36$ eV corresponding to an intrachain π – π^* transition, with two small shoulders at energies of 2.23 and 2.06 eV corresponding to interchain π – π stacking interactions and high planarity.^{31,32} The UV–vis spectra of PBTTT thin films depict a main absorption peak around 2.24 eV followed by a weak shoulder around 2.12 eV, which are linked to intra- and interchain π – π^* transitions, respectively.^{11,33} The intensity of the weak shoulder in PBTTT is stronger compared to P3HT, which is a manifestation of high crystallinity and more backbone planarity in the PBTTT thin film, as confirmed by GIWAXS analysis. On the other hand, UV–vis spectra of OD-PDPP2T-TT thin films show a strong absorption below 2 eV. Two peaks in absorption were observed around 1.66 and 1.50 eV, which are consistent with reported spectra for OD-PDPP2T-TT.²⁷

Next, Figure 2b shows the absorption spectra of the doped films. The data are shown until 0.5 eV due to the limitation of the ultraviolet–visible–near-infrared (UV–vis–NIR) spectrometer. The P_2 peak is clearly visible in the UV–vis–NIR spectra for all polymers. Fourier transform infrared spectroscopy (FTIR) measurements were performed below 0.5 eV to

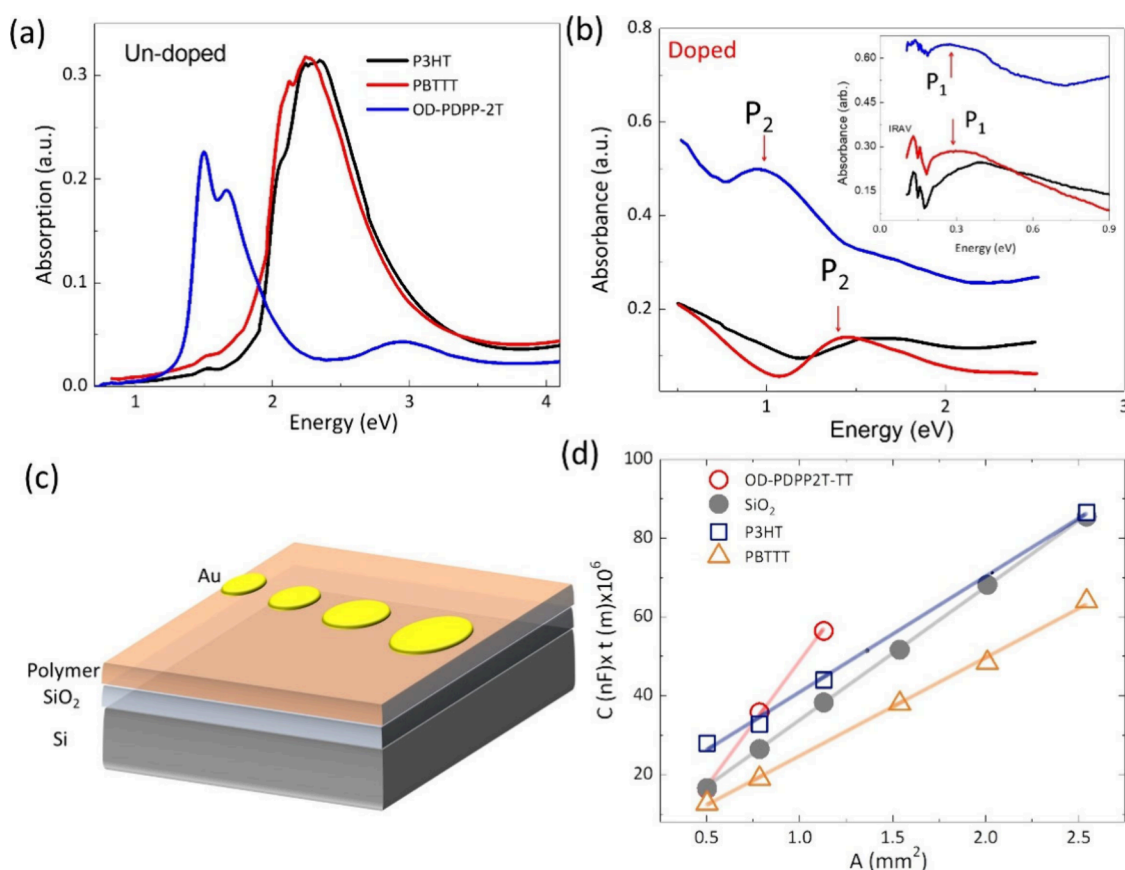


Figure 2. UV-vis spectra and dielectric measurement of P3HT, PBTTT-*C*₁₄, and OD-PDPP2T-TT polymers: UV-vis spectra of the (a) pristine and (b) doped P3HT, PBTTT-*C*₁₄, and OD-PDPP2T-TT polymer thin films with 0.03 M concentration of FeCl₃. The inset of panel b shows FTIR spectra of doped samples. (c) Schematic of the measurement geometry and the dielectric constant of different conjugated polymer films. (d) Capacitance (*C*) × thickness (*t*) of polymer thin films as a function of the area (*A*) for undoped P3HT (blue squares), PBTTT (orange triangles), OD-PDPP2T-TT (red circles), and SiO₂ (gray circles). Thin films of P3HT, PBTTT-*C*₁₂, and OD-PDPP2T-TT polymer were prepared by spin coating in dichlorobenzene solution.

observe the *P*₁ peak, as shown in the inset of Figure 2b. *P*₁ and *P*₂ absorption for P3HT is observed around 0.38 and 1.55 eV, which are close to those reported in the literature.³² In the doped PBTTT film, the *P*₂ peak is observed around 1.45 eV and the *P*₁ peak is found around 0.32 eV similar to those reported for PBTTT doped with FeCl₃ previously.³⁴ A red shift in the *P*₁ position in PBTTT compared to P3HT is a signature of more delocalized carriers linked to its high crystallinity.^{16,33} In the case of OD-PDPP2T-TT, the *P*₂ peak is observed around 1 eV and the *P*₁ peak was found around 0.27 eV. A red shift in *P*₁ for OD-PDPP2T-TT suggests that the carriers are even more delocalized compared to PBTTT. On the other hand, OD-PDPP2T-TT shows a relatively higher paracrystallinity (see Table 1). Therefore, higher crystallinity cannot be the reason for the red shift in the OD-PDPP2T-TT polymer thin film. An alternate factor that controls carrier delocalization is the coulombic interaction between polarons and counterions. For example, it has shown that, by choosing a larger size of dopant molecule, the average distance between the polaron and counterion can be increased, which decreases the coulombic interaction between polarons and counterions, giving rise to higher conductivity.^{32,35,36} In our case, the same dopant, namely, FeCl₃, was used. Therefore, we postulate that the dielectric environment of the polymer matrix must tune this coulombic interaction.

To support this hypothesis, we measured the static dielectric constant of all three polymers. For this, we devise a cross-plane configuration, as shown in the upper panel in Figure 2c, where the polymer film (on SiO₂) is sandwiched between patterned metal discs of different areas on the top and the highly conducting doped silicon substrate at the bottom. The capacitance (*C*) multiplied by the thickness (*t*) of each polymer thin film as a function of the area (*A*) for P3HT, PBTTT-*C*₁₂, and OD-PDPP2T-TT, respectively, is plotted in Figure 2d. The capacitance of SiO₂ as a function of the area was measured as a reference without any polymer films. The slope of *C* × *t* versus *A* represents the dielectric constant of the polymer (see the Supporting Information for more details about the chosen axis). The value of the static dielectric constant for SiO₂ was found to be 3.2, which is consistent with the reported value in the literature.³⁷ After measurement of the SiO₂ dielectric constant, polymer thin films were spin-coated on SiO₂. While the *C* × *t* versus *A* slope for P3HT and PBTTT looks similar to SiO₂, as evidenced in Figure 2d, a higher slope is observed for OD-PDPP2T-TT. The dielectric constant of each polymer was deduced using a series combination of capacitances from SiO₂ and the polymer (for more details, please see the methods of the Supporting Information). The resultant dielectric constant values are listed in Table 2. Higher dielectric constant values for OD-PDPP2T-TT were also

supported by density functional theory (DFT) calculations (see the [Supporting Information](#) for more details).

Table 2. Measured Static Dielectric Constant of Different Polymers, with SiO₂ Used as a Reference

	SiO ₂	P3HT	PBTBT	OD-PDPP2T-TT
dielectric constant (ϵ_s)	3.3	2.8	2.5	6.3

Armed with experimentally measured morphology, extent of delocalization, and dielectric constants of the polymers, we measure the thermoelectric transport properties of the polymer films with the aim to develop a modified charge transport model to fully understand their thermoelectric performance. [Figure 3a](#) shows the schematic of our vacuum de-doping experiment to generate a large range of Seebeck coefficient versus conductivity data (over 3 orders of magnitude change). All three polymers were first doped with a 0.03 M concentration of FeCl₃, and de-doping was done under high vacuum (for more details, please see the methods of the [Supporting Information](#)). In short, under vacuum, the dopants start to escape from the polymer matrix, which causes de-doping. In our previous work,³⁸ we have demonstrated that structural disorder results in a Gaussian tail in the density of states (DOS). Therefore, the total effective DOS can be written as a convolution function of the original DOS and the Gaussian broadening function, given by (for more details, see the methods of the [Supporting Information](#))

$$D_v(E) = \frac{1}{\sqrt{\pi}} \int D_v^0 \left((E_v - E) - \sigma_{\text{eff}} \frac{E}{w} \right) e^{-E^2/w^2} dE \quad (1)$$

where $D_v^0(E)$ is the DOS of the valence band without structural disorder ($g = 0$), w is the Gaussian broadening due to structural disorder, and σ_{eff} is the effective Gaussian broadening due to both the screening effect and the structural disorder [see [Figure S3](#) of the Supporting Information demonstrating the modified three-dimensional (3D) DOS with a Gaussian tail, in which the modified DOS around the Fermi level can penetrate the gap region]. Using the generalized Boltzmann transport equation, the electrical conductivity can then be described as

$$\sigma = \frac{2q^2}{3m_c^*} \int \tau(E) E D_v(E) \left(-\frac{\partial f}{\partial E} \right) dE \quad (2)$$

where $\tau(E)$ is the relaxation time and can be written as a contribution from acoustic phonons $\tau_{\text{AP}}(E)$ and impurity scattering $\tau_{\text{II}}(E)$ times, respectively, within Mathieson's rule framework

$$\frac{1}{\tau(E)} = \frac{1}{\tau_{\text{AP}}(E)} + \frac{1}{\tau_{\text{II}}(E)} \quad (3)$$

The Seebeck coefficient can be defined as

$$S = \frac{1}{eT\sigma} \int_0^\infty \sigma_E \left(-\frac{\partial f}{\partial E} \right) (E - E_F) dE \quad (4)$$

[Figure 3b](#) shows the measured Seebeck coefficient (S) as a function of electrical conductivity (σ) of P3HT, PBTBT-C₁₂, and OD-PDPP2T-TT thin films. As expected, S increases as σ decreases with de-doping for all three polymers. This *in situ* de-doping technique allows us to consider the same value of the Seebeck coefficient for all three polymers, therefore corresponding to the same doping level (horizontal black dashed

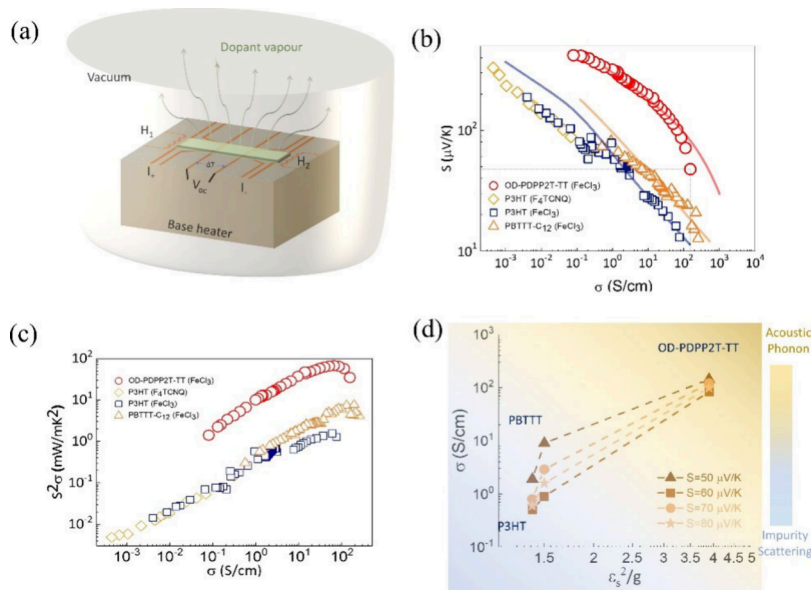


Figure 3. Measurement of the thermoelectric power factor ($S^2\sigma$) of dopant- and dielectric-controlled polymers: (a) Vacuum chamber de-dopes the polymeric thin films under continuous application of vacuum, while the patterned heaters and thin-film thermometers can measure $S^2\sigma$ *in situ*. (b) Measured Seebeck coefficient (S) as a function of conductivity (σ) and (c) power factor as a function of conductivity of P3HT (square), PBTBT (triangle), and OD-PDPP2T-TT (circle) thin films. A charge transport model using effective density of states (more details in the [Supporting Information](#)) was applied on the P3HT, PBTBT, and OD-PDPP2T-TT thin films, respectively, and fits well in the delocalized carrier regime ($\sigma > 1$ S/cm), while it does not capture the localized carrier regime (details in the [Supporting Information](#)). (d) Conductivity as a function of $\frac{\epsilon_s^2}{g}$, where ϵ_s is the static dielectric constant of a pure polymer matrix without any doping and g is the paracrystallinity. Conductivity values were taken at different Seebeck coefficient values.

line in Figure 3b). Here, we chose a Seebeck value of $50 \mu\text{V/K}$, which is the point at which the power factor peaks for the OD-PDPP2T-TT polymer, as seen in Figure 3c. At this value, OD-PDPP2T-TT shows >14 times higher conductivity compared to P3HT and PBTTT- C_{12} thin films (vertical dashed line in Figure 3b), which can be attributed to the higher dielectric constant with strong counterion screening. Solid color lines in Figure 3b show the calculated values from the charge transport model and illustrate that experimental data of S versus σ for all three polymers coincide well with our modified charge transport model for a conductivity higher than 1 S/cm , where the modified DOS model is accurate. Interestingly, for P3HT and PBTTT- C_{12} , the theoretically modeled solid lines show a near-linear trend in the experimental data, and OD-PDPP2T-TT sees a distinct slope change at higher conductivity values (close to 100 S/cm). We hypothesize that this slope change is caused by a change in the scattering of the mobile polarons from impurity to acoustic phonon limited, which is analogous to metallic transport ($S \propto \frac{1}{\sigma}$) in PBTTT- C_{12} by electrolyte gating, as discussed by Tanaka et al.²⁹ Note that, below a conductivity of 1 S/cm , we expect Fermi level pinning in localized states due to structural disorder, and hence, a hopping transport model is more relevant^{39,40} than our delocalized charge transport model, which is suitable to explain transport at higher doping regimes.

Next, we observe the maximum power factor in our films by plotting the power factor ($S^2\sigma$) as a function of the conductivity in Figure 3c. The power factor initially increases with conductivity for all polymers until it shows a peak at higher conductivity values and then drops. The optimal power factor value was found to be 1.58 , 7.8 , and $66 \mu\text{W m}^{-1} \text{ K}^{-2}$ for P3HT, PBTTT, and OD-PDPP2T-TT, respectively. A 9-fold enhancement in the power factor is therefore observed for OD-PDPP2T-TT compared to the PBTTT film, without any attempts to alter the morphology. To understand the higher power factor in OD-PDPP2T-TT compared to PBTTT and P3HT better, in Figure 3d, we plot conductivity as a function of $\frac{\epsilon_s^2}{g}$ at different Seebeck coefficient values, where ϵ_s is the static dielectric constant of a pure polymer and g is the paracrystallinity. Then, we consider the individual contributions to the relaxation times: $\tau_{\text{AP}}(E)$ and $\tau_{\text{II}}(E)$, which are expected to be proportional to the effective mass of the charge carriers (see eqs 10 and 13 of the Supporting Information for details). For ionized impurity-dominated scattering, the mobility of the carriers is proportional to the square of the static dielectric constant (ϵ_s) and square root of conductivity mass (m^*) (see eq 13 of the Supporting Information). However, $\tau_{\text{AP}}(E)$ does not depend upon the static dielectric environment (see eq 10 of the Supporting Information). Paracrystallinity, g , alters the degree of energetic disorder in the electronic structure significantly.¹² This structural disorder broadens the density of states, which will increase the effective mass of the carriers. Therefore, g can be correlated with m^* of conducting polymers.¹² For a Seebeck value of $75 \mu\text{V/K}$, which corresponds to a doping level where the charge transport is dominated by impurity scattering, OD-PDPP2T-TT shows 1 order of magnitude higher conductivity compared to P3HT and PBTTT due to a larger ratio of $\frac{\epsilon_s^2}{g}$, as shown in Figure 3d. A linear trend is observed between conductivity and $\frac{\epsilon_s^2}{g}$ in this regime. As the Seebeck coefficient drops from 75 to

$50 \mu\text{V/K}$, the linear trend starts to deviate. This deviation occurs as the scattering mechanism starts to switch from impurity scattering to acoustic phonon limited for OD-PDPP2T-TT; on the other hand, charge transport in both P3HT and PBTTT continues to be dominated by impurity scattering (details in Figure S4 of the Supporting Information). The deviation increases further as the Seebeck coefficient drops with increased doping. For the lowest Seebeck coefficient ($50 \mu\text{V/K}$), the conductivity value for OD-PDPP2T-TT is comparable to that of PBTTT. At this doping level, the dominant scattering mechanism is acoustic phonon scattering in OD-PDPP2T-TT (details in Figure S4 of the Supporting Information). Here, the effect of the dielectric constant and counterion screening on relaxation times diminishes, and only structural order dependence remains (see eq 6 of the Supporting Information). Therefore, as seen in Figure 3d, the power factor for OD-PDPP2T-TT drops sharply and drops to a value lower than that for PBTTT at the highest doping level. This is expected as, at high doping regimes, PBTTT is more structurally ordered in comparison to OD-PDPP2T-TT (see Table 1 and Figure 1c). Hence, in the optimal doping range, the dielectric constant of the polymer is a robust metric that determines the counterion screening and, therefore, controls charge transport toward achieving maximal thermoelectric performance. While we clearly see an enhanced performance due to the dielectric constant providing screening of mobile polarons, this is still far from the maximum potential that can be achieved in conducting polymers.

CONCLUSION

In summary, the present study reveals that the thermoelectric power factor in conducting polymers can be further improved by controlling the dielectric constant of the polymer without regulating the structural disorder. A 9-fold enhancement was observed in the power factor for OD-PDPP2T-TT compared to PBTTT. The maximum power factor is observed at the crossover point from impurity scattering to acoustic phonon scattering in OD-PDPP2T-TT. By leveraging the higher mobility of charge carriers arising from a higher dielectric constant, we can obtain a large electrical conductivity in these conducting polymers without sacrificing the Seebeck coefficient. Finally, we expect that building on this work and developing new methods to improve the dielectric constant of the polymer matrix and further control the morphology will provide a new avenue to augment organic thermoelectric materials.

MATERIALS AND METHODS

Materials. Electronic-grade P3HT was bought from Rieke metals (weight-average molecular weight M_w of $\sim 50\text{--}70 \text{ kg mol}^{-1}$); PBTTT- (C_{14}) (weight-average molecular weight M_w of $\sim 40\text{--}80 \text{ kg mol}^{-1}$) was purchased from Sigma-Aldrich; and OD-PDPP2T-TT (weight-average molecular weight M_w of $\sim 250\text{--}300 \text{ kg mol}^{-1}$) was purchased from Ossila and used as received. FeCl_3 was purchased from Sigma-Aldrich. The solvents, 1,2-dichlorobenzene (purity of >99.9%) and nitromethane (99.9%, extra dry, over molecular sieve), were purchased from Sigma-Aldrich.

Thin-Film Fabrication. P3HT, PBTTT, and OD-PDPP2T-TT powders were dissolved in dichlorobenzene (DCB) with a 10 mg/mL concentration and stirred overnight at 70°C on a hot plate. The polymer solutions were spin-coated (1000 rpm for 90 s) on pre-cleaned (acetone, isopropanol, and UV-ozone cleaner) and pre-patterned quartz substrates for electrical and thermoelectric measurements. Pre-cleaned n-doped silicon substrates were used for GIWAXS

measurement. For UV–vis measurement, films were spun-coated on pre-cleaned 1 in. quartz substrates. Sequential doping was performed in a glovebox ($O_2 < 1$ ppm and $H_2O < 1$ ppm) using 0.03 M $FeCl_3$ solution in nitromethane by dipping the polymer film in dopant solution for 5 s.

GIWAXS Measurement. GIWAXS measurements were performed in the Lawrence Berkeley National Laboratory. The energy of the incident beam was 10 keV, and a Pilatus 2M area detector was used. The X-ray scattering data were taken at incidence angles of $0.14 \pm 0.005^\circ$ with 1–5 s exposure times. The samples were kept under a helium environment during X-ray exposure to minimize air scattering and sample degradation. Samples were doped before measurement. Before loading, sample resistance was measured. It was in the $\sim 1000 \Omega$ range. After measurement, sample resistance was measured again, and it was found in the $\sim 2000 \Omega$ range. Therefore, before and after measurement, the films were still heavily doped.

UV–Vis Measurement. A Shimadzu UV 3600 spectrophotometer was used to measure the absorption spectra of pristine and doped polymer samples. Quartz substrate was used as a reference. The absorption was measured from 4 to 0.5 eV. The films were doped just before the absorption measurement to avoid any degradation issue.

FTIR Measurement. A Bruker VERTEX 80v spectrophotometer was used to measure the FTIR absorption spectra of doped polymer samples. A KBr disc was used as the substrate. The polymer films were spin-coated on a KBr disc. The absorption was measured from 0.9 down to 0.1 eV. The films were doped just before the absorption measurement to avoid any degradation issue.

Electrical and Thermoelectric Characterization. The electrical conductivity and Seebeck coefficient of polymer thin films were measured using a shadow mask technique to pattern the heater and electrodes to measure the in-plane thermoelectric properties. The polymer film is spin-coated on a pre-patterned quartz substrate. After sequential doping, the films were loaded in a high-vacuum ($\sim 10^{-6}$ Torr) cryostat for electrical and thermoelectric measurements. The film thickness was measured before performing de-doping measurement with a KLA Tencor profilometer. The vacuum chamber with a heating stage was used to de-dope the polymer films; the dopant slowly evaporates out of the film and de-dopes the film sequentially in time at room temperature. After reaching vacuum de-doping saturation at room temperature, the film was annealed in vacuum at $50^\circ C$ for 30 min and then brought back to room temperature for measurement. This process was repeated multiple times to remove the dopant further. The measurement was performed until sample resistance reached the $1 M\Omega$ range. Above $1 M\Omega$, the open-circuit voltage was noisy and the signal-to-noise ratio was higher. The electrical conductivity and Seebeck coefficient were calculated as a function of this de-doping.

Dielectric Measurement. The metal insulator semiconductor (MIS) device structure was used to measure the low-frequency dielectric constant of pristine polymer materials, as shown in Figure S4 of the Supporting Information. P3HT, PBTTT, and OD-PDPP2T-TT were dissolved in dichlorobenzene (DCB) with 10 mg/mL concentration and stirred overnight at $70^\circ C$ on a hot plate. The polymer solutions were then spun-casted onto pre-cleaned SiO_2 (300 nm)/Si substrates at 1000 rpm in a N_2 -filled glovebox. The spin-coated films were annealed at $120^\circ C$ for 10 min to remove the solvent residue. The thickness of the films was measured using a KLA Tencor profilometer. A thickness of $33 \pm 5 \text{ nm}$ was found for P3HT and PBTTT, and a thickness of $63 \pm 5 \text{ nm}$ was found for OD-PDPP2T-TT. Cr (10 nm) and Au (100 nm) were deposited as electrodes with different surface areas using a shadow mask. Device fabrication was performed in a N_2 -filled glovebox. Keithley 4200 with the CVU option was used to measure the capacitance of each area size device. An alternating current (AC) voltage of 100 mV and 10 kHz frequency and a direct current (DC) voltage sweep from +30 to -10 V were applied for measurement. The capacitance of SiO_2 without a polymer film was measured as a function of different electrode dot sizes. After that, the polymer film capacitance with different electrode dot sizes was measured. A +30 V was applied to the structure to ensure a fully depleted device. The depleted polymer layer acts as a

capacitor in series with the SiO_2 layer, and hence, the total capacitance is

$$\frac{1}{C_{\text{exp}}} = \frac{1}{C_{SiO_2}} + \frac{1}{C_{\text{polymer}}} \quad (5)$$

where C_{SiO_2} and C_{polymer} are the capacitances of SiO_2 and the polymer, respectively. After extraction of the capacitance of the polymer thin film, the dielectric constant of the polymer was calculated using the following equation:

$$C_{\text{polymer}} = \frac{\epsilon_0 \epsilon_r A}{t} \quad (6)$$

where A is the electrode area, ϵ_r is the polymer dielectric constant, ϵ_0 is the vacuum permittivity, and t is the polymer film thickness.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.4c20823>.

Details about electrical characterization and theoretical calculation (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Pawan Kumar – Institute of Materials Research and Engineering, Agency for Science Technology and Research, Singapore 138634, Singapore; orcid.org/0000-0001-9772-5638; Email: pawankumar@imre.a-star.edu.sg

Kedar Hippalgaonkar – Institute of Materials Research and Engineering, Agency for Science Technology and Research, Singapore 138634, Singapore; School of Materials Research and Engineering, Nanyang Technological University, Singapore 639798, Singapore; orcid.org/0000-0002-1270-9047; Email: kedar@ntu.edu.sg

Authors

Anas Abutaha – Qatar Environment and Energy Research Institute (QEERI), Hamad Bin Khalifa University (HBKU), Doha, Qatar; orcid.org/0000-0002-8923-5417

Gang Wu – Institute of High Performance Computing, Agency for Science Technology and Research, Singapore 138632, Singapore; orcid.org/0000-0002-0055-0200

Madeleine P. Gordon – Molecular Foundry, Lawrence Berkeley National Laboratory, Berkeley, California 94720, United States; Applied Science and Technology Graduate Group, University of California, Berkeley, Berkeley, California 94720, United States

Jose Recatala-Gomez – School of Materials Research and Engineering, Nanyang Technological University, Singapore 639798, Singapore; orcid.org/0000-0003-3560-2768

Jeffrey J. Urban – Molecular Foundry, Lawrence Berkeley National Laboratory, Berkeley, California 94720, United States; orcid.org/0000-0003-4909-2869

Complete contact information is available at: <https://pubs.acs.org/doi/10.1021/acsami.4c20823>

Author Contributions

[†]Pawan Kumar, Anas Abutaha, and Gang Wu contributed equally to this work.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

Work at the Molecular Foundry was supported by the Office of Science, Office of Basic Energy Sciences, of the US Department of Energy under contract number DE-AC02-05CH11231. Pawan Kumar and Kedar Hippalgaonkar acknowledge funding from the Materials Generative Design and Testing Framework (MAT-GDT) Program at A*STAR, provided through the AME Programmatic Fund (Grant M24N4b0034).

REFERENCES

- (1) Haddad, C.; Périllon, C.; Danlos, A.; François, M. X.; Descombes, G. Some Efficient Solutions to Recover Low and Medium Waste Heat: Competitiveness of the Thermoacoustic Technology. *Energy Procedia* **2014**, *50*, 1056–1069.
- (2) Shi, X. L.; Zou, J.; Chen, Z. G. Advanced Thermoelectric Design: From Materials and Structures to Devices. *Chem. Rev.* **2020**, *120* (15), 7399–7515.
- (3) Snyder, G.; Toberer, E. Complex Thermoelectric Materials. *Nat. Mater.* **2008**, *7*, 105–114.
- (4) Bubnova, O.; Khan, Z. U.; Malti, A.; Braun, S.; Fahlman, M.; Berggren, M.; Crispin, X. Optimization of the Thermoelectric Figure of Merit in the Conducting Polymer Poly(3,4-Ethylenedioxythiophene). *Nat. Mater.* **2011**, *10* (6), 429–433.
- (5) Kim, G. H.; Shao, L.; Zhang, K.; Pipe, K. P. Engineered Doping of Organic Semiconductors for Enhanced Thermoelectric Efficiency. *Nat. Mater.* **2013**, *12* (8), 719–723.
- (6) Bubnova, O.; Khan, Z. U.; Wang, H.; Braun, S.; Evans, D. R.; Fabretto, M.; Hojati-Talemi, P.; Dagnelund, D.; Arlin, J. B.; Geerts, Y. H.; Desbief, S.; Breiby, D. W.; Andreassen, J. W.; Lazzaroni, R.; Chen, W. M.; Zozoulenko, I.; Fahlman, M.; Murphy, P. J.; Berggren, M.; Crispin, X. Semi-metallic Polymers. *Nat. Mater.* **2014**, *13* (2), 190–194.
- (7) Vijayakumar, V.; Zhong, Y.; Untilova, V.; Bahri, M.; Herrmann, L.; Biniek, L.; Leclerc, N.; Brinkmann, M. Bringing Conducting Polymers to High Order: Toward Conductivities beyond 10^5 S cm^{-1} and Thermoelectric Power Factors of $2 \text{ mW m}^{-1} \text{ K}^{-2}$. *Adv. Ener. Mater.* **2019**, *9* (24), 1900266.
- (8) Untilova, V.; Hynynen, J.; Hofmann, A. I.; Scheunemann, D.; Zhang, Y.; Barlow, S.; Kemerink, M.; Marder, S. R.; Biniek, L.; Müller, C.; Brinkmann, M. High Thermoelectric Power Factor of Poly(3-hexylthiophene) through In-Plane Alignment and Doping with a Molybdenum Dithiolenes Complex. *Macromolecules* **2020**, *53* (15), 6314–6321.
- (9) Glauddell, A. M.; Cochran, J. E.; Patel, S. N.; Chabiny, M. L. Impact of the Doping Method on Conductivity and Thermopower in Semiconducting Polythiophenes. *Adv. Ener. Mater.* **2015**, *5*, 1401072.
- (10) Lim, E.; Peterson, K. A.; Su, G. M.; Chabiny, M. L. Thermoelectric Properties of Poly(3-hexylthiophene) (P3HT) Doped with 2,3,5,6-Tetrafluoro-7,7,8,8-tetracyanoquinodimethane (F4TCNQ) by Vapor-Phase Infiltration. *Chem. Mater.* **2018**, *30* (3), 998–1010.
- (11) Patel, S. N.; Glauddell, A. M.; Peterson, K. A.; Thomas, E. M.; O'Hara, K. A.; Lim, E.; Chabiny, M. L. Morphology Controls the Thermoelectric Power Factor of a Doped Semiconducting Polymer. *Sci. Adv.* **2017**, *3*, No. e1700434.
- (12) Noriega, R.; Rivnay, J.; Vandewal, K.; Koch, F. P. V.; Stingelin, N.; Smith, P.; Toney, M. F.; Salleo, A. A General Relationship Between Disorder, Aggregation and Charge Transport in Conjugated Polymers. *Nat. Mater.* **2013**, *12* (11), 1038–1044.
- (13) Jacobs, I. E.; Aasen, E. W.; Oliveira, J. L.; Fonseca, T. N.; Roehling, J. D.; Li, J.; Zhang, G.; Augustine, M. P.; Mascal, M.; Moule, A. J. Comparison of Solution-Mixed and Sequentially Processed P3HT:F4TCNQ Films: Effect of Doping-Induced Aggregation on Film Morphology. *J. Mater. Chem. C* **2016**, *4*, 3454–3466.
- (14) Scholes, D. T.; Hawks, S. A.; Yee, P. Y.; Wu, H.; Lindemuth, J. R.; Tolbert, S. H.; Schwartz, B. J. Overcoming Film Quality Issues for Conjugated Polymers Doped with F₄TCNQ by Solution Sequential Processing: Hall Effect, Structural, and Optical Measurements. *J. Phys. Chem. Lett.* **2015**, *6* (23), 4786–4793.
- (15) Scholes, D. T.; Yee, P. Y.; Lindemuth, J. R.; Kang, H.; Onorato, J.; Ghosh, R.; Luscombe, C. K.; Spano, F. C.; Tolbert, S. H.; Schwartz, B. J. The Effects of Crystallinity on Charge Transport and the Structure of Sequentially Processed F₄TCNQ-Doped Conjugated Polymer Films. *Adv. Funct. Mater.* **2017**, *27* (44), 1702654.
- (16) Vijayakumar, V.; Durand, P.; Zeng, H.; Untilova, V.; Herrmann, L.; Algayer, P.; Leclerc, N.; Brinkmann, M. Influence of Dopant Size and Doping Method on the Structure and Thermoelectric Properties of PBTTT Films Doped with F₆TCNNQ and F₄TCNQ. *J. Mater. Chem. C* **2020**, *8*, 16470–16482.
- (17) Aubry, T. J.; Winchell, K. J.; Salamat, C. Z.; Basile, V. M.; Lindemuth, J. R.; Stauber, J. M.; Axtell, J. C.; Kubena, R. M.; Phan, M. D.; Bird, M. J.; Spokoyny, A. M.; Tolbert, S. H.; Schwartz, B. J. Tunable Dopants with Intrinsic Counterion Separation Reveal the Effects of Electron Affinity on Dopant Intercalation and Free Carrier Production in Sequentially Doped Conjugated Polymer Films. *Adv. Funct. Mater.* **2020**, *30* (28), 2001800.
- (18) Brebels, J.; Manca, J. V.; Lutsen, L.; Vanderzande, D.; Maes, W. High Dielectric Constant Conjugated Materials for Organic Photovoltaics. *J. Mater. Chem. A* **2017**, *5*, 24037–24050.
- (19) Cho, N.; Schlenker, C. W.; Knesting, K. M.; Koelsch, P.; Yip, H.-L.; Ginger, D. S.; Jen, A. K.-Y. High-Dielectric Constant Side-Chain Polymers Show Reduced Non-Geminate Recombination in Heterojunction Solar Cells. *Adv. Energy Mater.* **2014**, *4* (10), 1301857.
- (20) Brebels, J.; Douvogianni, E.; Devisscher, D.; Thiruvallur Eachambadi, R.; Manca, J.; Lutsen, L.; Vanderzande, D.; Hummelen, J. C.; Maes, W. An Effective Strategy to Enhance the Dielectric Constant of Organic Semiconductors—CPDTPD-based Low Bandgap Polymers Bearing oligo(ethylene glycol) side chain. *J. Mater. Chem. C* **2018**, *6*, 500–511.
- (21) Wang, C.; Zhang, Z.; Pejić, S.; Li, R.; Fukuto, M.; Zhu, L.; Sauvé, G. High Dielectric Constant Semiconducting Poly(3-alkylthiophene)s from Side Chain Modification with Polar Sulfinyl and Sulfonyl Groups. *Macromolecules* **2018**, *51* (22), 9368–9381.
- (22) Marszałek, T.; Li, M.; Pisula, W. Design Directed Self-assembly of Donor-acceptor Polymers. *Chem. Commun.* **2016**, *52*, 10938–10947.
- (23) Yiu, A. T.; Beaujuge, P. M.; Lee, O. P.; Woo, C. H.; Toney, M. F.; Fréchet, J. M. J. Side-Chain Tunability of Furan-Containing Low-Band-Gap Polymers Provides Control of Structural Order in Efficient Solar Cells. *J. Am. Chem. Soc.* **2012**, *134* (4), 2180–2185.
- (24) Zhang, G.; Clarke, T. M.; Mozer, A. J. Bimolecular Recombination in a Low Bandgap Polymer:PCBM Blend Solar Cell with a High Dielectric Constant. *J. Phys. Chem. C* **2016**, *120* (13), 7033–7043.
- (25) Sirringhaus, H.; Brown, P. J.; Friend, R. H.; Nielsen, M. M.; Bechgaard, K.; Langeveld-Voss, B. M. W.; Spiering, A. J. H.; Janssen, R. A. J.; Meijer, E. W.; Herwig, P.; de Leeuw, D. M. Two-dimensional Charge Transport in Self-organized, High-mobility Conjugated Polymers. *Nature* **1999**, *401*, 685–688.
- (26) McCulloch, I.; Heeney, M.; Bailey, C.; Genevicius, K.; Macdonald, I.; Shkunov, M.; Sparrowe, D.; Tierney, S.; Wagner, R.; Zhang, W.; Chabiny, M. L.; Kline, R. J.; McGehee, M. D.; Toney, M. F. Liquid-Crystalline Semiconducting Polymers with High Charge-Carrier Mobility. *Nat. Mater.* **2006**, *5* (4), 328–333.
- (27) Zhang, X.; Richter, L. J.; DeLongchamp, D. M.; Kline, R. J.; Hammond, M. R.; McCulloch, I.; Heeney, M.; Ashraf, R. S.; Smith, J. N.; Anthopoulos, T. D.; Schroeder, B.; Geerts, Y. H.; Fischer, D. A.; Toney, M. F. Molecular Packing of High-Mobility Diketopyrrolopyrrole Polymer Semiconductors with Branched Alkyl Side Chains. *J. Am. Chem. Soc.* **2011**, *133*, 15073.
- (28) Rivnay, J.; Noriega, R.; Kline, R. J.; Salleo, A.; Toney, M. F. Quantitative Analysis of Lattice Disorder and Crystallite Size in Organic Semiconductor Thin Films. *Phys. Rev. B* **2011**, *84*, 045203.

(29) Tanaka, H.; Kanahashi, K.; Takekoshi, N.; Mada, H.; Ito, H.; Shimoi, Y.; Ohta, H.; Takenobu, T. Thermoelectric Properties of a Semicrystalline Polymer Doped Beyond the Insulator-to-Metal Transition by Electrolyte Gating. *Sci. Adv.* **2020**, *6*, No. eaay8065.

(30) Ghosh, R.; Pochas, C. M.; Spano, F. C. Polaron Delocalization in Conjugated Polymer Films. *J. Phys. Chem. C* **2016**, *120* (21), 11394–11406.

(31) Rasmussen, S. C.; Straw, B. D.; Hutchison, J. E. Tuning the Extent of Conjugation in Processable Polythiophenes through Control of Side Chain Density and Regioregularity. In *Semiconducting Polymers*; Hsieh, B. R., Wei, Y., Eds.; American Chemical Society (ACS): Washington, D.C., 1999; ACS Symposium Series, Vol. 735, Chapter 21, pp 347366, DOI: 10.1021/bk-1999-0735.ch021.

(32) Salleo, A.; Kline, R. J.; DeLongchamp, D. M.; Chabinyc, M. L. Microstructural Characterization and Charge Transport in Thin Films of Conjugated Polymers. *Adv. Mater.* **2010**, *22* (34), 3812–3838.

(33) Cochran, J. E.; Junk, M. J. N.; Glaudell, A. M.; Miller, P. L.; Cowart, J. S.; Toney, M. F.; Hawker, C. J.; Chmelka, B. F.; Chabinyc, M. L. Molecular Interactions and Ordering in Electrically Doped Polymers: Blends of PBTTT and F4TCNQ. *Macromolecules* **2014**, *47*, 6836.

(34) Jacobs, I. E.; Lin, Y.; Huang, Y.; Ren, X.; Simatos, D.; Chen, C.; Tjhe, D.; Statz, M.; Lai, L.; Finn, P. A.; Neal, W. G.; D'Avino, G.; Lemaur, V.; Fratini, S.; Beljonne, D.; Strzalka, J.; Nielsen, C. B.; Barlow, S.; Marder, S. R.; McCulloch, I.; Sirringhaus, H. High-Efficiency Ion-Exchange Doping of Conducting Polymers. *Adv. Mater.* **2022**, *34*, 2102988.

(35) Aubry, T. J.; Axtell, J. C.; Basile, V. M.; Winchell, K. J.; Lindemuth, J. R.; Porter, T. M.; Liu, J.-Y.; Alexandrova, A. N.; Kubiak, C. P.; Tolbert, S. H.; Spokoiny, A. M.; Schwartz, B. J. Dodecaborane-Based Dopants Designed to Shield Anion Electrostatics Lead to Increased Carrier Mobility in a Doped Conjugated Polymer. *Adv. Mater.* **2019**, *31*, 1805647.

(36) Thomas, E. M.; Peterson, K. A.; Balzer, H.; Rawlings, D.; Stingelin, N.; Segalman, R. A.; Chabinyc, M. L. Effects of Counter-Ion Size on Delocalization of Carriers and Stability of Doped Semiconducting Polymers. *Adv. Electron. Mater.* **2020**, *6*, 2000595.

(37) Lim, S. W.; Shimogaki, Y.; Nakano, Y.; Tada, K.; Hiroshi Komiyama, H. K. Preparation of Low-Dielectric-Constant F-Doped SiO₂ Films by Plasma-Enhanced Chemical Vapor Deposition. *Jpn. J. Appl. Phys.* **1996**, *35*, 1468.

(38) Abutaha, A.; Kumar, P.; Yildirim, E.; Shi, W.; Yang, S.-W.; Wu, G.; Hippalgaonkar, K. Correlating Charge and Thermoelectric Transport to Paracrystallinity in Conducting Polymers. *Nat. Commun.* **2020**, *11*, 1737.

(39) Zuo, G.; Abdalla, H.; Kemerink, M. Impact of doping on the density of states and the mobility in organic semiconductors. *Phys. Rev.* **2016**, *93*, 235203.

(40) Abdalla, H.; Zuo, G.; Kemerink, M. Range and energetics of charge hopping in organic semiconductors. *Phys. Rev.* **2017**, *96*, No. 241202(R).



CAS INSIGHTS™

EXPLORE THE INNOVATIONS SHAPING TOMORROW

Discover the latest scientific research and trends with CAS Insights. Subscribe for email updates on new articles, reports, and webinars at the intersection of science and innovation.

Subscribe today

CAS
A division of the
American Chemical Society